

STAGE 4 TECHNOLOGY MANDATORY

Programmable Light | 25-period teaching program

Engineering technologies and systems; Digital and communication technologies | Syllabus-aligned network review copy v2.0

Course	Stage 4 Technology Mandatory	Syllabus	Technology 7-8 Syllabus (2023)
Delivery	10 weeks 25 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	10 Site/theory 5 Project/folio 10 Practical	Implementation	Current and mandatory from 2026
Official source	NSW Curriculum official page verified 28 Aug 2026	Focus	Engineering technologies and systems; Digital and communication technologies
Status	Local review copy matching Word, PDF and manifest v2.0	Approval	Teacher to confirm before delivery
Complete Stage 4 outcomes appear below. Period rows identify the outcomes directly supported by each lesson; this does not claim that every outcome is formally assessed in this unit.			

Official Stage 4 outcomes

Outcome	Exact official wording	Outcome	Exact official wording
TE4-SDP-01	explains relationships between sustainability, design and production	TE4-DES-01	communicates and evaluates design ideas and solutions
TE4-PDP-01	describes the practices and processes of designers and producers	TE4-SAF-01	selects and safely uses tools, materials, technologies and processes
TE4-MSC-01	explains how materials, systems and components contribute to solutions	TE4-DIG-01	demonstrates technological literacy to safely interact in digital environments
TE4-PPM-01	applies processes in the planning, management and production of projects	TE4-DIG-02	uses data and digital systems to code, design and produce projects

What students are learning

Learning focus	In plain English	Learning focus	In plain English
Sustainability, design and production	Explain how design and production choices affect sustainability.	Design communication and evaluation	Communicate ideas and judge solutions against criteria.
Designers and producers	Describe how people investigate, plan, make and improve solutions.	Safe practical work	Choose and use approved tools, materials, technologies and processes safely.
Materials, systems and components	Explain how parts, materials and systems affect how a solution works.	Safe digital literacy	Work safely and responsibly in digital environments.
Project planning and production	Plan, manage and produce the project in a clear sequence.	Data, coding and digital systems	Use data or digital systems where they support the unit's design and production.

Teacher-to-confirm register

When	Decision needed	Confirm by
Before the unit	Timetable, lesson order, class supports and approved adjustments	Teacher / faculty
Before practical work	Activity, materials, equipment, supervision and current risk controls	Teacher / faculty
Before assessment	Task authority, marks, dates, conditions and submission method	Teacher / faculty
Before classroom issue	Current links, program version and any local sequence changes	Teacher / faculty

Teaching program | Periods 1-10

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
1	<u>Site/theory</u>	Module 1 - What makes a programmable lamp?; Workshop safety: hazards.	Teach named section(s); students complete each 10-question check and saved response.	Investigate materials and production processes; develop. Outcomes: TE4-SAF-01, TE4-PDP-01	check + response
2	<u>Project/folio</u>	Module 1 - folio evidence	Review responses; organise source, planning, feedback and evaluation evidence.	Plan, document, communicate and evaluate design and production Outcomes: TE4-DES-01, TE4-PPM-01	Check the current folio and evaluation evidence
3	Practical	Module 1 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
4	Practical	Module 1 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
5	<u>Site/theory</u>	Module 2 - Image quality, resolution and design research.	Teach named section(s); students complete each 10-question check and saved response.	Investigate materials and production processes; develop. Outcomes: TE4-DES-01, TE4-PDP-01	check + response
6	<u>Project/folio</u>	Module 2 - folio evidence	Review responses; organise source, planning, feedback and evaluation evidence.	Plan, document, communicate and evaluate design and production Outcomes: TE4-DES-01, TE4-PPM-01	Check the current folio and evaluation evidence
7	Practical	Module 2 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
8	Practical	Module 2 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
9	<u>Site/theory</u>	Module 3 - Hardwood, softwood and a reasoned material choice.	Teach named section(s); students complete each 10-question check and saved response.	Investigate materials and production processes; develop. Outcomes: TE4-MSC-01, TE4-PDP-01	check + response
10	<u>Site/theory</u>	Module 3 - Planning timber production from drawing to evidence	Teach named section(s); students complete each 10-question check and saved response.	Investigate materials and production processes; develop. Outcomes: TE4-MSC-01, TE4-PDP-01	check + response
Teacher to confirm before use: Periods 1-10 practical activities, materials, equipment, supervision and current risk controls.					

Teaching program | Periods 11-20

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
11	Project/folio	Module 3 - folio evidence	Review responses; organise source, planning, feedback and evaluation evidence.	Plan, document, communicate and evaluate design and production Outcomes: TE4-DES-01, TE4-PPM-01	Check the current folio and evaluation evidence
12	Practical	Module 3 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
13	Practical	Module 3 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
14	Site/theory	Module 4 - Coding language: algorithms, sequencing and syntax	Teach named section(s); students complete each 10-question check and saved response.	Develop and test digital solutions using data, algorithms, systems and components Outcomes: TE4-DIG-02, TE4-PDP-01	check + response
15	Site/theory	Module 4 - Variables, loops and conditionals for light behaviour	Teach named section(s); students complete each 10-question check and saved response.	Develop and test digital solutions using data, algorithms, systems and components Outcomes: TE4-DIG-02, TE4-PDP-01	check + response
16	Site/theory	Module 4 - Connecting code, controller and light as a system	Teach named section(s); students complete each 10-question check and saved response.	Develop and test digital solutions using data, algorithms, systems and components Outcomes: TE4-DIG-02, TE4-PDP-01	check + response
17	Project/folio	Module 4 - folio evidence	Review responses; organise source, planning, feedback and evaluation evidence.	Plan, document, communicate and evaluate design and production Outcomes: TE4-DES-01, TE4-PPM-01	Check the current folio and evaluation evidence
18	Practical	Module 4 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
19	Practical	Module 4 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
20	Site/theory	Module 5 - Integrating design, materials, electronics and code	Teach named section(s); students complete each 10-question check and saved response.	Develop and test digital solutions using data, algorithms, systems and components Outcomes: TE4-DIG-02, TE4-PDP-01	check + response

Teacher to confirm before use: Periods 11-20 practical activities, materials, equipment, supervision and current risk controls.

Teaching program | Periods 21-25

5 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
21	<u>Site/theory</u>	Module 5 - Testing, debugging and improving the solution	Teach named section(s); students complete each 10-question check and saved response.	Develop and test digital solutions using data, algorithms, systems and components Outcomes: TE4-DIG-02, TE4-PDP-01	check + response
22	<u>Site/theory</u>	Module 5 - Evaluating quality, sustainability and learning	Teach named section(s); students complete each 10-question check and saved response.	Investigate materials and production processes; develop. Outcomes: TE4-SDP-01, TE4-PDP-01	check + response
23	<u>Project/folio</u>	Module 5 - folio evidence	Review responses; organise source, planning, feedback and evaluation evidence.	Plan, document, communicate and evaluate design and production Outcomes: TE4-DES-01, TE4-PPM-01	Check the current folio and evaluation evidence
24	Practical	Module 5 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
25	Practical	Module 5 - authorised practical	Follow current teacher demonstration/local procedure; record genuine observations or progress.	Safe selection and use of tools, materials and production processes Outcomes: TE4-SAF-01, TE4-PPM-01	Observe the approved practical and check the authentic record
Teacher to confirm before use: Periods 21-25 practical activities, materials, equipment, supervision and current risk controls.					

Evidence, feedback and support

Evidence	Look for	Respond / support
Theory and module work	Understanding and use of key ideas	Return to the named section and model one stronger response
Project and folio	Design decisions, planning and refinement	Conference against criteria and give one clear next step
Practical observation	Safety, workflow, skill, quality and clean-down	Pause, demonstrate and recheck before continuing
Product and evaluation	Quality and evidence-based judgement	Compare the result with the original criteria